

Multiple Linear Regression

Theory Questions and Solutions

TTK4260

Note. The solutions below are intended as model answers; alternative correct explanations/derivations earn full credit.

Question 1 (*Model statement and notation*)

Consider the multiple linear regression (MLR) model

$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_p x_p + \varepsilon, \quad \text{equivalently } \mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}.$$

- (a) Define the dimensions of $\mathbf{y}$, $\mathbf{X}$, $\boldsymbol{\theta}$ and $\boldsymbol{\varepsilon}$ for a dataset with m observations and p predictors.
[4 points]
- (b) Explain why $\mathbf{X}$ typically contains a column of ones. [2 points]
- (c) State the ordinary least squares (OLS) objective in both summation form and matrix norm form. [4 points]

Solution.

- (a) With intercept, $\mathbf{X} \in \mathbb{R}^{m \times (p+1)}$, $\boldsymbol{\theta} \in \mathbb{R}^{(p+1) \times 1}$, and $\mathbf{y}, \boldsymbol{\varepsilon} \in \mathbb{R}^{m \times 1}$.
- (b) A column of ones allows a free intercept θ_0 , i.e., the baseline level of y when all predictors are zero (in the model scale).
- (c) OLS minimizes the sum of squared residuals:

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^m (y_i - \theta_0 - \sum_{j=1}^p \theta_j x_{ij})^2 \quad \Leftrightarrow \quad \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2.$$

Question 2 (*Interpretation: “holding others fixed”*)

Suppose the fitted model for house price is

$$\widehat{\text{Price}} = \theta_0 + \theta_1 \cdot \text{Size} + \theta_2 \cdot \text{Age} + \theta_3 \cdot \text{Bedrooms}.$$

- (a) In words, interpret θ_1 and explicitly include the phrase “holding others fixed” (or equivalent). [3 points]
- (b) Give one situation where interpreting θ_1 as a causal effect would be questionable. [3 points]
- (c) If the model includes a predictor that is highly correlated with Size, what can happen to the sign and stability of $\hat{\theta}_1$? Explain briefly. [4 points]

Solution.

- (a) Holding **Age** and **Bedrooms** fixed, increasing **Size** by one unit changes the *expected* price by θ_1 units (on the response scale).

- (b) Causal interpretation is questionable if there are omitted confounders (e.g., neighborhood/location quality) that affect both size and price, or if size is endogenous (measurement/selection issues).
- (c) With strong correlation (multicollinearity), $\hat{\theta}_1$ can have large standard error, become unstable across samples, and even flip sign (because the model struggles to uniquely assign “credit” among correlated predictors).

Question 3 (*Confounding example*)

You observe that ice-cream sales y increase with temperature x_1 . You later add a binary variable x_2 indicating weekend (1) vs weekday (0).

- (a) Explain qualitatively how omitting x_2 can bias the estimated slope of temperature. [5 points]
- (b) Write the two models (with and without x_2) and describe what changes in the interpretation of the temperature coefficient when x_2 is included. [5 points]

Solution.

- (a) If weekends tend to be warmer *and* weekends have higher sales due to leisure/foot traffic, then x_2 is positively correlated with both y and x_1 . Omitting x_2 attributes part of the weekend effect to temperature, typically biasing the temperature slope upward (omitted-variable bias).
- (b) Without x_2 : $y = \theta_0 + \theta_1 x_1 + \varepsilon$. With x_2 : $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \varepsilon$. Including x_2 makes θ_1 the effect of temperature *holding weekend/weekday fixed*, i.e., comparing days with the same weekend status.

Question 4 (*Orthogonality vs. correlation*)

- (a) What does it mean for predictors to be orthogonal/uncorrelated in the context of regression design? [3 points]
- (b) State one practical advantage of orthogonal predictors for *interpretation* and one for *numerical stability*. [4 points]
- (c) Explain why high correlation can make $\mathbf{X}^\top \mathbf{X}$ nearly singular and what that implies for coefficient variance. [5 points]

Solution.

- (a) Predictors are orthogonal if their centered columns have zero inner product (equivalently, sample correlation ≈ 0 after centering). In matrix terms, columns of $\mathbf{X}$ (often after mean-centering) satisfy $\mathbf{x}_j^\top \mathbf{x}_k = 0$ for $j \neq k$.
- (b) Interpretation: each coefficient reflects a “clean” effect without shared variation, so “holding others fixed” is less contentious. Numerical stability: $\mathbf{X}^\top \mathbf{X}$ is closer to diagonal, easier to invert, and estimates have smaller variance.
- (c) If columns are nearly linear combinations of each other, $\mathbf{X}^\top \mathbf{X}$ has very small eigenvalues (ill-conditioned). Then $(\mathbf{X}^\top \mathbf{X})^{-1}$ has large entries, inflating $\text{Var}(\hat{\boldsymbol{\theta}})$ and making coefficients sensitive to noise.

Question 5 (“Who gets the credit?”)

Assume $x_2 \approx x_1$ in your dataset. Consider three models:

$$\hat{y} = 5 + 10x_1 + 0x_2, \quad \hat{y} = 5 + 0x_1 + 10x_2, \quad \hat{y} = 5 + 5x_1 + 5x_2.$$

- (a) Explain why these models can yield very similar predictions if $x_1 \approx x_2$. [4 points]
- (b) Which of the three models is easiest to *interpret* as a statement about the effect of x_1 alone? Justify. [3 points]
- (c) In one sentence, summarize the difference between “good prediction” and “good interpretation” in this setting. [3 points]

Solution.

- (a) If $x_1 \approx x_2$, then $10x_1 \approx 10x_2$ and $5x_1 + 5x_2 \approx 10x_1 \approx 10x_2$, so the linear predictors are close for most observations.
- (b) The first model, because it assigns all weight to x_1 and none to x_2 (least ambiguous “credit” assignment).
- (c) Prediction depends on the combined linear predictor, while interpretation depends on uniquely attributing effects to specific predictors, which correlation prevents.

Question 6 (Short computation: OLS by normal equations)

You are given $m = 3$ observations with two predictors and an intercept:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

(Here the predictors are columns 2 and 3.)

- (a) Compute $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$. [6 points]
- (b) Solve the normal equations to obtain $\hat{\boldsymbol{\theta}}$. [6 points]
- (c) Interpret the coefficient associated with the *second* predictor (column 3). [3 points]

Solution.

- (a) Let columns be $\mathbf{c}_1 = [1, 1, 1]^\top$, $\mathbf{c}_2 = [0, 1, 2]^\top$, $\mathbf{c}_3 = [1, 1, 1]^\top$. Then

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 5 & 3 \\ 3 & 3 & 3 \end{bmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}.$$

- (b) Since $\mathbf{c}_1 = \mathbf{c}_3$, $\mathbf{X}$ is rank-deficient, so $\mathbf{X}^\top \mathbf{X}$ is singular and the OLS minimizer is *not unique*. Only the sum $\alpha = \theta_0 + \theta_3$ is identifiable:

$$\hat{y} = \theta_0 + \theta_1 x + \theta_3 \cdot 1 = (\theta_0 + \theta_3) + \theta_1 x = \alpha + \theta_1 x.$$

Regressing y on $x = [0, 1, 2]$ with intercept gives

$$\hat{\theta}_1 = \frac{1}{2}, \quad \hat{\alpha} = \frac{7}{6}.$$

Thus any $(\theta_0, \theta_1, \theta_3)$ with $\theta_1 = \frac{1}{2}$ and $\theta_0 + \theta_3 = \frac{7}{6}$ is an OLS solution (e.g., $\hat{\boldsymbol{\theta}} = (\frac{7}{6}, \frac{1}{2}, 0)^\top$).

- (c) Column 3 is perfectly collinear with the intercept, so its individual coefficient is not separately interpretable; only $\theta_0 + \theta_3$ matters.

Question 7 (*Perfect multicollinearity*)

- (a) Give an example of a deterministic linear relationship among columns of $\mathbf{X}$ that creates *perfect* multicollinearity. [3 points]
- (b) What happens to the OLS solution when $\mathbf{X}^\top \mathbf{X}$ is singular? Mention both existence and uniqueness. [4 points]
- (c) Name one practical way to resolve perfect multicollinearity in a model. [3 points]

Solution.

- (a) Example: with intercept, including both `Height` and `HeightMeters=Height/100`, or including all k dummies and an intercept.
- (b) Rank deficiency $\Rightarrow \mathbf{X}^\top \mathbf{X}$ singular. Minimizers exist but are not unique; coefficients are not identifiable, although the fitted values $\mathbf{X}\hat{\boldsymbol{\theta}}$ are unique.
- (c) Drop a redundant column (or re-encode the categorical variable), or use regularization (e.g., ridge) if the goal is stable prediction rather than unique interpretation.

Question 8 (*Categorical variables: dummy encoding*)

A variable `Color` has three categories: `Red`, `Green`, `Blue`. You want to include it in a regression model with an intercept.

- (a) Construct a valid dummy/indicator encoding using $k - 1$ dummies. Specify the baseline. [5 points]
- (b) Write the regression model that includes the dummies and explain the meaning of the intercept under your encoding. [4 points]
- (c) Interpret the coefficient of the `IsRed` dummy. [3 points]

Solution.

- (a) Choose baseline `Blue`. Use D_{Red} and D_{Green} as indicators. Blue corresponds to $(0, 0)$.
- (b) $y = \theta_0 + \theta_1 D_{\text{Red}} + \theta_2 D_{\text{Green}} + \varepsilon$. θ_0 is the expected response for Blue.
- (c) θ_1 is the difference (Red minus Blue) in expected response.

Question 9 (*The dummy variable trap*)

Suppose you instead include *all* k dummies for a categorical variable with k levels, *and* you include an intercept.

- (a) Prove (briefly) that the design matrix has linearly dependent columns. [5 points]
- (b) Explain what you would observe if you attempt to compute $(\mathbf{X}^\top \mathbf{X})^{-1}$. [3 points]
- (c) Give two fixes and explain which one you prefer in practice and why. [4 points]

Solution.

- (a) Row-wise, exactly one dummy equals 1, so $D_1 + \dots + D_k = \mathbf{1}$. With an intercept column $\mathbf{1}$, columns are dependent.
- (b) The inverse does not exist (singular matrix); software may drop a column, return NaNs, or warn about rank deficiency.
- (c) Fix by dropping one dummy (preferred, clear baseline interpretation) or dropping the intercept (then each dummy corresponds to a category mean).

Question 10 (*Interactions with a categorical variable*)

You model test score y as a function of study hours x and group membership D_B (0 = Group A, 1 = Group B):

$$y = \theta_0 + \theta_1 x + \theta_2 D_B + \theta_3 (x \cdot D_B) + \varepsilon.$$

- (a) Write the fitted line (intercept and slope) for Group A and for Group B. [4 points]
- (b) Interpret θ_2 and θ_3 in words. [4 points]
- (c) In what real-world situation would adding the interaction term be justified? [3 points]

Solution.

- (a) Group A: $\hat{y} = \theta_0 + \theta_1 x$. Group B: $\hat{y} = (\theta_0 + \theta_2) + (\theta_1 + \theta_3)x$.
- (b) θ_2 is the baseline shift for Group B vs A at $x = 0$. θ_3 is the additional slope for Group B (difference in marginal effect of x).
- (c) When the effect of studying plausibly differs by group (e.g., different teaching methods or prior preparation).

Question 11 (*Overfitting intuition*)

- (a) Explain, using a training vs. test error argument, why adding more predictors can *increase* test error. [5 points]
- (b) Give one symptom in the fitted coefficients that suggests overfitting or instability in OLS with many predictors. [3 points]

Solution.

- (a) More predictors reduce training SSE but can increase variance by fitting noise, so test error can increase even though training error decreases.
- (b) Large/erratic coefficients, high variance across resamples, or very wide confidence intervals.

Question 12 (*Regularization: general idea*)

Consider the regularized objective

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \cdot \text{Penalty}(\boldsymbol{\theta}), \quad \lambda > 0.$$

- (a) What is the role of λ ? Describe what happens as $\lambda \rightarrow 0$ and as λ becomes very large. [4 points]

- (b) List three motivations for regularization (e.g., prediction, stability, interpretability) and connect each to the penalty term. [6 points]

Solution.

- (a) λ controls shrinkage/complexity: $\lambda \rightarrow 0$ yields (approximately) OLS; large λ forces parameters toward the penalty-minimizing values (often toward 0).
- (b) Motivation: reduce variance/improve prediction; handle multicollinearity via shrinkage and better conditioning; obtain sparse/simple models (for $L1/L0$) to aid interpretability.

Question 13 (*Ridge regression (L2)*)

Ridge regression solves

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2.$$

- (a) Derive the ridge closed-form solution $\hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$. [7 points]
- (b) Why does adding $\lambda \mathbf{I}$ improve conditioning / invertibility? [4 points]
- (c) State (and briefly justify) whether ridge typically produces *exact* zeros in coefficients. [2 points]

Solution.

- (a) Set gradient to zero:

$$-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) + 2\lambda\boldsymbol{\theta} = 0 \Rightarrow (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})\boldsymbol{\theta} = \mathbf{X}^\top \mathbf{y}.$$

$$\text{Hence } \hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}.$$

- (b) Adds λ to all eigenvalues, making the matrix positive definite and better conditioned.
- (c) No: ridge shrinks continuously; coefficients are rarely exactly zero.

Question 14 (*Lasso (L1)*)

Lasso solves

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1.$$

- (a) Explain geometrically (in words) why L1 regularization can produce exact zeros. [5 points]
- (b) Give one advantage and one caveat of lasso when predictors are highly correlated. [4 points]
- (c) Why is there generally no simple closed-form solution for lasso? [3 points]

Solution.

- (a) The $L1$ feasible set has corners aligned with axes; optimum often hits a corner $\Rightarrow$ some coordinates exactly 0.
- (b) Advantage: sparse selection; caveat: instability and arbitrary choice among correlated features.

- (c) Nondifferentiable at 0, so requires iterative convex optimization methods rather than a simple linear system solve.

Question 15 (*L0 vs L1 vs L2: comparison table*)

Fill in a table comparing $L0$, $L1$ (lasso), and $L2$ (ridge) along the following axes:

- sparsity (does it set coefficients to zero?),
- computational difficulty,
- stability under correlated predictors,
- interpretability.

Then, in two sentences, explain when you would prefer each method.

[10 points]

Solution.

Penalty	Sparsity	Computation	Correlation stability	Interpretability
$L0$	Yes	Hard (subset search)	Often unstable	High (few vars)
$L1$	Often yes	Convex (iterative)	Can be unstable	High-ish (sparse)
$L2$	No	Closed form / fast	Stable, groups vars	Medium

Use $L0$ (or approximations) for best-subset style selection; use lasso for sparse models; use ridge for stable prediction under multicollinearity.

Question 16 (*Feature scaling: interpretation*)

- (a) Explain how changing the units of a predictor (e.g., meters to kilometers) changes the numerical value of its regression coefficient. [4 points]
- (b) Suppose θ_1 is the coefficient for **YearsExperience**. If you standardize **YearsExperience** to have mean 0 and standard deviation 1, how does the interpretation of the coefficient change? [4 points]

Solution.

- (a) If $x' = cx$, then $\theta' = \theta/c$ to keep $\theta x = \theta' x'$. So changing units rescales the coefficient inversely.
- (b) The coefficient becomes the expected change in y per *one standard deviation* increase in experience (holding other predictors fixed).

Question 17 (*Feature scaling: regularization*)

- (a) Explain why scaling is optional for OLS but important for ridge and lasso. [5 points]
- (b) Consider two predictors $x_1 \in [0, 1]$ and $x_2 \in [0, 1000]$. Without scaling, which coefficient tends to be penalized more by an $L2$ penalty, and why? [5 points]
- (c) State the critical rule about fitting a scaler when doing train/test evaluation. [2 points]

Solution.

- (a) OLS optimum is unaffected in a meaningful way by rescaling (you can rescale coefficients and keep the same fit), but ridge/lasso penalize coefficient sizes, so scale changes how strongly each feature is effectively penalized.
- (b) x_2 can use a much smaller coefficient to have the same impact on $\hat{y}$, so it is often penalized less; x_1 may need a larger coefficient and is penalized more. Scaling fixes this mismatch.
- (c) Fit the scaler on training data only; apply the same transform to validation/test data (no leakage).

Question 18 (*Standardization vs min-max normalization*)

- (a) Write the formulas for z-score standardization and min-max normalization. [4 points]
- (b) Give one scenario where min-max normalization can be problematic, and explain why. [3 points]
- (c) If your goal is to compare effect sizes across predictors in a linear model, which scaling method is usually more convenient, and why? [3 points]

Solution.

- (a) z-score: $x' = (x - \mu)/\sigma$. Min-max: $x' = (x - \min)/(\max - \min)$.
- (b) Sensitive to outliers (they set min/max) and to distribution shift (test values outside training range).
- (c) Standardization, because “one unit” corresponds to one standard deviation, making coefficients comparable.

Question 19 (*Concept check: multiple choice*)

Circle **all** statements that are true.

- (a) If predictors are highly correlated, prediction can still be good even if individual coefficients are hard to interpret.
- (b) With an intercept in the model, a k -level categorical variable should be encoded with k dummies (one-hot) to avoid information loss.
- (c) Ridge regression can be viewed as OLS with an added constraint $\|\theta\|_2^2 \leq C$ for some C .
- (d) Lasso always selects the same variable from a set of highly correlated predictors, regardless of small data changes.
- (e) Scaling features is essential before using lasso or ridge if you want the penalty to treat predictors fairly.

[5 points]

Solution. True: (a), (c), (e). False: (b), (d).

Question 20 (*Open-ended synthesis*)

You are given a dataset with: (i) 50 predictors, many of which are strongly correlated, (ii) only 200 observations, (iii) a need for an interpretable model that highlights a small set of important predictors.

- (a) Propose a modeling approach (OLS / ridge / lasso / elastic net / other) and justify your choice in terms of stability and sparsity. *[6 points]*
- (b) List two concrete preprocessing steps you would apply before fitting, and explain why. *[4 points]*

Solution.

- (a) Elastic net is often best here: it provides sparsity (via $L1$) and improved stability under correlated predictors (via $L2$). Lasso alone can be too unstable with strong correlations; ridge is stable but not sparse.
- (b) Standardize predictors (fair penalty, numerical conditioning) and tune hyperparameters using cross-validation; additionally handle missing values/outliers consistently and remove constant features.